Replication V2: the Joux–Vitse sieve against a strong rho

Thresholds fixed before data; acceptance rule A4 failed, every decision withheld. Weak class only; the cover's Jacobian has genus 3. Study `research/jv_sieve_strong_rho_v2_20261020`, seeds 17–32, run 2026-10-20. Every number is copied from `summary.json`, computed by `examples/jv_sieve_study_v2.rs`.

Outcome: the pre-registered acceptance rule A4 failed, so every decision is withheld. Everything else in the acceptance list passed. The tables below are descriptive; they are not claims.

Scope: a weak class only. Curves with full rational 2-torsion whose isogeny class contains a weak curve, never prime order; the cover's Jacobian has **genus 3**. Class: measurement of an existing algorithm against a stronger reference. Protocol with numeric thresholds, committed before any run: `PROTOCOL.md` (one amendment, below). Numbers: `TABLES.md`, `summary.json` raw runs: `raw/` report: `REPORT.pdf`.

What was run

New instances (seeds 17–32, disjoint from the first study's 1–16) at $p \in \{251, 503, 1009, 1511, 1823\}$ (ℓ = 2^45.8 … 2^63.0), on the corrected producer of PR #2281 (DP table keeps `y`, every `hash_x` evaluation counted, option validation). The producer's commit and binary hash were recorded before the first run (`raw/PROVENANCE.json`: commit `d688f3c2b`, `jv_cover` SHA-256 `1e63ff73…da848`). Arms, samples, prices and the §5a band are the first study's; 16 runs per size for the strong rho (R3) and the sieve (dense and sparse), a 2×2 rho factorial, a plain-rho control, and A/A repeats at $p = 1009$.

The acceptance rules

rule	requirement	result
A1	complete seed inventories, every run correct and uncapped	pass
A2	hash-scope completeness per run (2 hash evaluations per step plain, 4 with negation)	pass (the engine emits exactly 2.0000 and 4.0000)
A3	plain rho reproduces the ledger's S ≈ 1.3 (interval contains 1.3, point in [1.0, 1.6])	pass at all five sizes (points 1.06, 1.33, 1.19, 1.21, 1.28)
A4	**rho A/A at p = 1009: ratio of means in [0.85, 1.15] and interval contains 1**	**FAIL**
A5	sieve A/A at p = 1009: identical counts and digests	pass (16/16)
A6	dense and sparse FGLM identical descent digests and outcomes	pass (80/80)
A7	strong-rho exponent point in [0.45, 0.55], interval contains 0.5	pass (0.511 [0.483, 0.540])

What failed. R0 first run over repeat: 0.792 [0.572, 1.073]; R3: 0.729 [0.516, 1.035]. Both intervals contain 1; both point estimates are outside [0.85, 1.15]. Under the protocol a failed acceptance rule withholds D1–D6, so **no K1/K2 decision, no exponent classification, no crossover and no acceleration claim is made from this study**, and the first study's qualified status is unchanged.

Why I think the rule was mis-specified, and why I did not change it. The ±0.15 window has no power calculation behind it. With a per-run spread of about 50 % (the A/A arms have coefficients of variation 0.39–0.64) and 16 runs, the standard error of a ratio of two arm means is about 0.18, so the window is 0.85 standard errors wide: a sound instrument would fail it about 40 % of the time per arm, about 64 % for two. The failure is therefore weak evidence of an instrument defect; the intervals contain 1 and no difference is detected. That is an argument about the rule, not a licence to relax it after seeing the data: the protocol says a failed rule is reported as failed, and a changed threshold is a new protocol version. The cheapest honest next step is a V3 with a rule sized to the sample (for example, the interval of the ratio must contain 1, with an equivalence margin of about two standard errors, or enough A/A runs, roughly 85 per arm for ±0.15), fixed before a further campaign.

Amendment (before the final analysis)

A dry run of the analyzer on partial data showed that A1 only inspected arms that had finished runs, so a scheduled arm with none would have passed. The analyzer now fails A1 for any absent scheduled arm (`missing_arms`, unit-tested). No threshold, window, sample or rule changed; the producer was not rebuilt. Logged at the end of `PROTOCOL.md`.

Descriptive results (not decisions)

Sieve/R3 ratios, paired by instance seed, 95 % bootstrap (below 1 is the sieve cheaper):

p	S, harness	S, tuned	S^+, harness	S^+, tuned
251	45.8 [5.2, 137]	110 [13, 332]	86.4 [11, 249]	170 [22, 496]
503	0.647 [0.45, 0.93]	1.56 [1.1, 2.2]	1.78 [1.2, 2.6]	3.5 [2.3, 5.1]
1009	0.0289 [0.023, 0.036]	0.0695 [0.055, 0.086]	0.0759 [0.063, 0.093]	0.149 [0.12, 0.18]
1511	0.00738 [0.0055, 0.0099]	0.0177 [0.013, 0.024]	0.025 [0.019, 0.032]	0.0493 [0.038, 0.064]
1823	0.00404 [0.0033, 0.0051]	0.00971 [0.0079, 0.012]	0.014 [0.011, 0.018]	0.0275 [0.022, 0.036]

For orientation only: the first study (seeds 1–16, earlier producer) read 0.066 [0.053, 0.083], 0.027 [0.022, 0.036] and 0.015 [0.012, 0.019] for the S^+ harness ratio at $p = 1009, 1511, 1823$ against 0.076, 0.025 and 0.014 here; its fitted sieve exponents were −0.122 and 0.052 (post hoc, $p \geq 1009$) against −0.152 [−0.208, −0.107] and 0.042 [−0.020, 0.103] here; its strong-rho exponent 0.508 against 0.511. Different seeds

and a different producer, so this is a description of agreement, not a test. At `p = 503` the sieve is below the strong rho in `S` (0.647 [0.447, 0.932]) and above it in `S⁺` (1.78 [1.18, 2.61]); `p = 251` is far above.

In this tower batching shows no saving in the harness price (R1 against R0, R3 against R2 in TABLES.md T3), negation lowers `S` (R2 against R0), and sparse FGLM leaves the descent rows identical (80/80) with the linear stage 2.08–2.20× cheaper and `S` 1.006–1.135× cheaper. These are descriptions of this sample; the decision rules D5 and D6 were not evaluated.

Limits

Weak class only; `p ≤ 1823` (ℓ < 2^63). The isogeny walk to reach the class was not re-measured. The tuned price is a cost model that credits rho only. The sieve's F4 additions are not emitted by the harness; the band uses a labelled upper bound. Wall time is an observation only. One instance at `p = 251` (seed 22) needed `m = 11` and took 31 minutes per sieve arm: a heavy tail of the sieve at small `p`, deterministic and included.

Figures

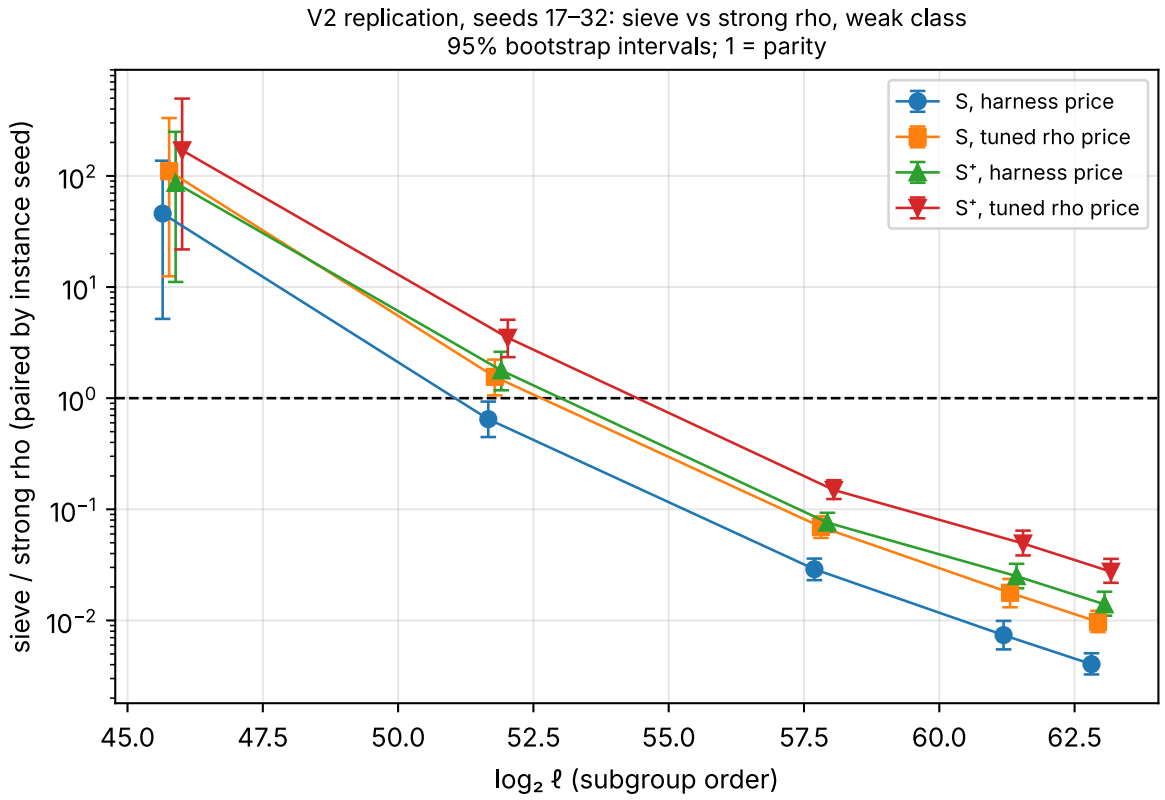


Figure 1. Sieve over strong rho (R3), paired by seed, four prices; 1 is parity. Study jv_sieve_strong_rho_v2_20261020, seeds 17–32, 95% bootstrap intervals; source summary.json. Descriptive: acceptance rule A4 failed, decisions withheld.

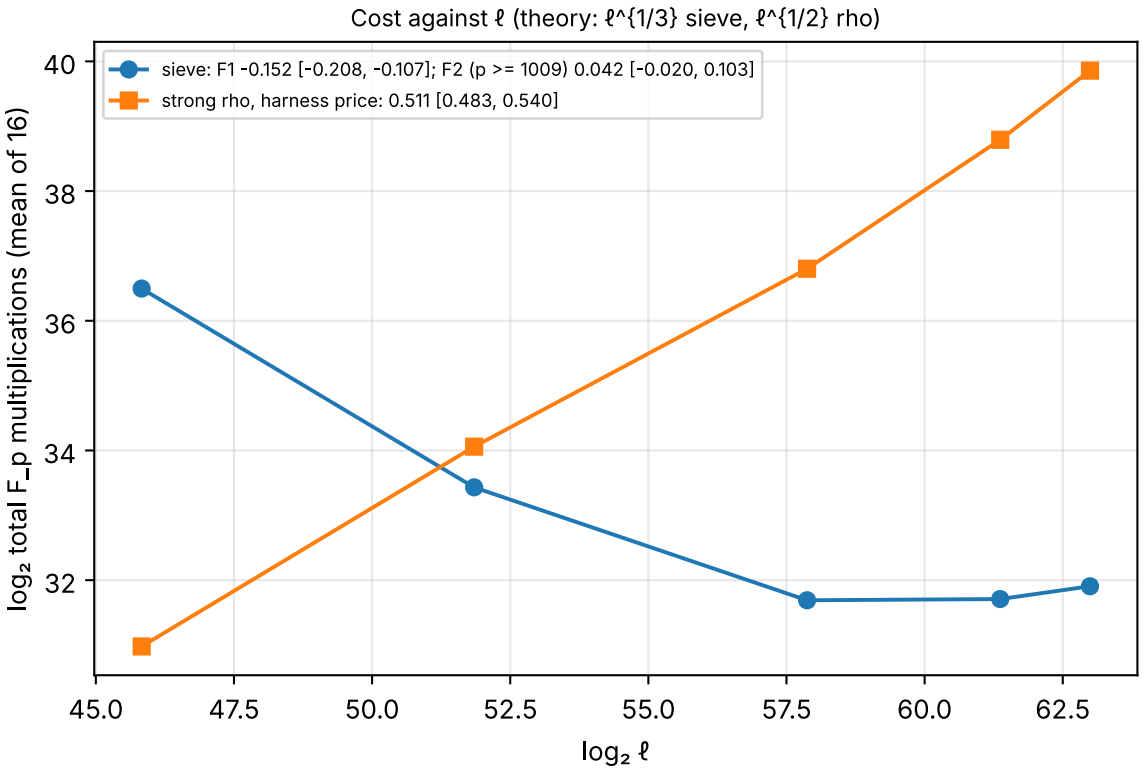


Figure 2. Mean total F_p multiplications against ℓ; fitted slopes in the legend. Study jv_sieve_strong_rho_v2_20261020, seeds 17–32, 95% bootstrap intervals; source summary.json. Descriptive: acceptance rule A4 failed, decisions withheld.

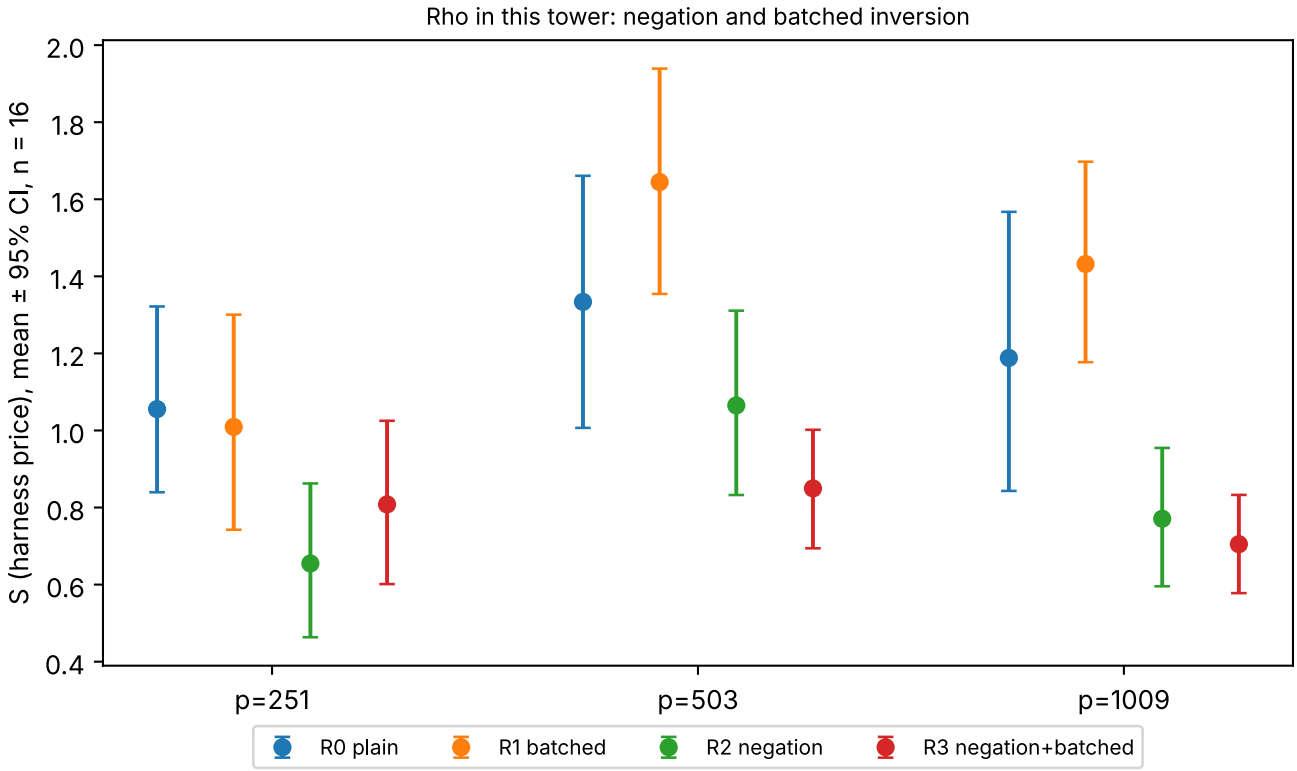


Figure 3. Rho factorial at p = 251, 503, 1009, harness price. Study jv_sieve_strong_rho_v2_20261020, seeds 17–32, 95% bootstrap intervals; source summary.json. Descriptive: acceptance rule A4 failed, decisions withheld.

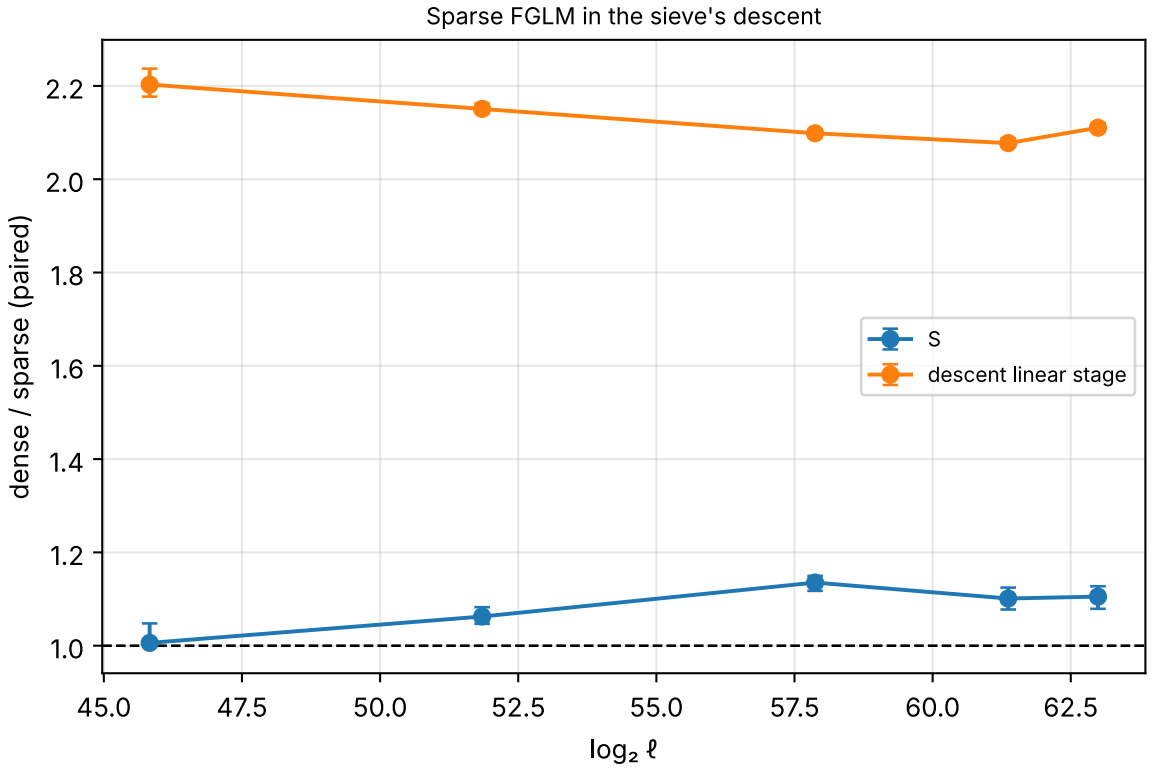


Figure 4. Dense over sparse FGLM in the sieve's descent, paired. Study jv_sieve_strong_rho_v2_20261020, seeds 17–32, 95% bootstrap intervals; source summary.json. Descriptive: acceptance rule A4 failed, decisions withheld.

T0. Acceptance rules (PROTOCOL.md): every decision is withheld unless

all pass

rule	requirement	result
A1	complete seed inventories, every run correct and uncapped	**pass**
A2	hash-scope completeness per run (2 per step plain, 4 with negation)	**pass**
A3	plain rho reproduces the ledger's S ≈ 1.3 (interval contains 1.3, point in [1.0, 1.6])	**pass**
A4	rho A/A at p = 1009 (ratio in [0.85, 1.15], interval contains 1)	**FAIL**
A5	sieve A/A at p = 1009 identical counts and digests	**pass**
A6	dense and sparse FGLM identical descent digests and outcomes	**pass**
A7	strong-rho exponent point in [0.45, 0.55], interval contains 0.5	**pass**

All pass: **False**. Failed: ['A4: rho A/A at p=1009']

T1. Sieve against the strong rho (R3), paired by instance seed

S, S⁺ in units of 331·√ℓ F_p multiplications. Ratios are sieve/rho: below 1 is the sieve cheaper. 95% bootstrap over instances.

p	log₂ℓ	n	sieve S	sieve S⁺	rho S (harness / tuned)	rho S⁺ (harness / tuned)	S ratio, harness	S ratio, tuned	S⁺ ratio, harness	S⁺ ratio, tuned
251	45.8	16	37.04 [4.55, 91.9]	82.92 [11.6, 204]	0.808 / 0.336	0.960 / 0.488	45.83 [5.17, 137]	110.2 [12.5, 332]	86.41 [11.1, 249]	170.1 [21.8, 496]
503	51.8	16	0.55 [0.411, 0.709]	1.794 [1.29, 2.35]	0.850 / 0.354	1.009 / 0.513	0.6473 [0.447, 0.932]	1.556 [1.06, 2.22]	1.778 [1.18, 2.61]	3.498 [2.34, 5.08]
1009	57.9	16	0.02038 [0.0171, 0.024]	0.06351 [0.0556, 0.072]	0.705 / 0.293	0.837 / 0.426	0.02891 [0.023, 0.036]	0.06948 [0.0553, 0.0864]	0.07586 [0.0626, 0.093]	0.1492 [0.124, 0.183]
1511	61.4	16	0.006148 [0.00552, 0.0069]	0.02478 [0.0233, 0.0262]	0.833 / 0.347	0.989 / 0.503	0.00738 [0.00549, 0.00991]	0.01773 [0.0131, 0.0236]	0.02504 [0.0195, 0.0323]	0.04926 [0.0385, 0.0643]

1823	63.0	16	0.004013 [0.00341, 0.00473]	0.01648 [0.0149, 0.0181]	0.993 / 0.413	1.180 / 0.600	0.00404 [0.00327, 0.00507]	0.00971 [0.00786, 0.0122]	0.01397 [0.011, 0.0181]	0.02748 [0.0218, 0.0358]

T2. §5a band: sieve/rho with native work priced at w GAE per

operation on every arm

p	w	A, harness	A, tuned	B, harness	B, tuned
251	0.0	45.8 [5.2, 133]	110 [12, 333]	45.8 [5.2, 140]	110 [13, 321]
	0.1	272 [41, 767]	296 [44, 863]	30.8 [5.1, 89]	39.4 [6.6, 109]
	1.0	304 [46, 858]	307 [46, 895]	23.2 [5.1, 64]	24 [5.3, 63]
503	0.0	0.647 [0.44, 0.93]	1.56 [1.1, 2.3]	0.647 [0.45, 0.93]	1.56 [1.1, 2.2]
	0.1	7.48 [5, 11]	8.13 [5.5, 12]	0.804 [0.55, 1.2]	1.03 [0.69, 1.5]
	1.0	8.45 [5.7, 12]	8.53 [5.7, 13]	0.883 [0.59, 1.3]	0.913 [0.61, 1.3]
1009	0.0	0.0289 [0.023, 0.036]	0.0695 [0.056, 0.087]	0.0289 [0.023, 0.036]	0.0695 [0.055, 0.087]
	0.1	0.353 [0.29, 0.43]	0.384 [0.32, 0.47]	0.0314 [0.026, 0.038]	0.0401 [0.033, 0.049]
	1.0	0.4 [0.33, 0.49]	0.403 [0.33, 0.49]	0.0326 [0.027, 0.04]	0.0337 [0.028, 0.041]
1511	0.0	0.00738 [0.0055, 0.0099]	0.0177 [0.013, 0.024]	0.00738 [0.0055, 0.0099]	0.0177 [0.013, 0.024]
	0.1	0.119 [0.093, 0.15]	0.129 [0.1, 0.17]	0.0103 [0.008, 0.013]	0.0132 [0.01, 0.017]
	1.0	0.135 [0.1, 0.17]	0.136 [0.11, 0.18]	0.0118 [0.0093, 0.015]	0.0122 [0.0096, 0.016]
1823	0.0	0.00404 [0.0033, 0.0051]	0.00971 [0.0079, 0.012]	0.00404 [0.0033, 0.0051]	0.00971 [0.0079, 0.012]
	0.1	0.067 [0.053, 0.086]	0.073 [0.058, 0.095]	0.00576 [0.0046, 0.0075]	0.00735 [0.0059, 0.0095]
	1.0	0.076 [0.06, 0.098]	0.0767 [0.061, 0.1]	0.00662 [0.0052, 0.0088]	0.00684 [0.0054, 0.009]

T3. Rho factorial and controls (S, harness price)

p	arm	n	steps/$\sqrt{\ell}$	S harness	S tuned	S^+ harness	wall s (1 core)
251	R0 plain	16	1.05 [0.84, 1.3]	1.06 [0.84, 1.3]	0.408 [0.33, 0.51]	1.21 [0.97, 1.5]	3
251	R1 batched	16	1.01 [0.75, 1.3]	1.01 [0.74, 1.3]	0.42 [0.31, 0.54]	1.15 [0.85, 1.5]	2
251	R2 negation	16	0.648 [0.46, 0.86]	0.655 [0.46, 0.86]	0.253 [0.18, 0.33]	0.777 [0.55, 1]	2
251	R3 negation + batched	16	0.803 [0.59, 1]	0.808 [0.6, 1]	0.336 [0.25, 0.42]	0.96 [0.71, 1.2]	2
503	R0 plain	16	1.33 [1, 1.7]	1.33 [1, 1.7]	0.516 [0.39, 0.65]	1.52 [1.2, 1.9]	33
503	R1 batched	16	1.65 [1.4, 1.9]	1.64 [1.4, 1.9]	0.684 [0.56, 0.81]	1.88 [1.6, 2.2]	31
503	R2 negation	16	1.06 [0.83, 1.3]	1.06 [0.83, 1.3]	0.412 [0.32, 0.51]	1.26 [0.98, 1.5]	28
503	R3 negation + batched	16	0.849 [0.7, 1]	0.85 [0.69, 1]	0.354 [0.29, 0.42]	1.01 [0.83, 1.2]	17
1009	R0 plain	16	1.19 [0.84, 1.6]	1.19 [0.84, 1.6]	0.46 [0.32, 0.61]	1.36 [0.96, 1.8]	223
1009	R1 batched	16	1.44 [1.2, 1.7]	1.43 [1.2, 1.7]	0.596 [0.49, 0.7]	1.64 [1.3, 1.9]	220
1009	R2 negation	16	0.769 [0.59, 0.95]	0.771 [0.6, 0.95]	0.298 [0.23, 0.37]	0.915 [0.71, 1.1]	167
1009	R3 negation + batched	16	0.705 [0.58, 0.83]	0.705 [0.58, 0.83]	0.293 [0.24, 0.35]	0.837 [0.68, 0.99]	121
1009	R0 A/A repeat	16	1.5 [1.1, 1.9]	1.5 [1.1, 1.9]	0.58 [0.43, 0.74]	1.71 [1.3, 2.2]	301
1009	R3 A/A repeat	16	0.966 [0.72, 1.2]	0.966 [0.72, 1.2]	0.402 [0.3, 0.52]	1.15 [0.86, 1.5]	159
1511	R0 plain	8	1.21 [0.9, 1.5]	1.21 [0.9, 1.5]	0.468 [0.35, 0.58]	1.38 [1, 1.7]	804
1511	R3 negation + batched	16	0.833 [0.65, 1.1]	0.833 [0.65, 1.1]	0.347 [0.27, 0.44]	0.989 [0.77, 1.3]	466
1823	R0 plain	4	1.28 [0.53, 2.1]	1.28 [0.53, 2.1]	0.495 [0.2, 0.81]	1.46 [0.61, 2.4]	1476
1823	R3 negation + batched	16	0.993 [0.74, 1.3]	0.993 [0.75, 1.3]	0.413 [0.31, 0.53]	1.18 [0.88, 1.5]	957

T4. Dense against sparse FGLM in the sieve's descent

p	pairs	identical digest	identical outcome	S dense/sparse	linear stage dense/sparse	descent muls dense/sparse
251	16	16	16	1.006 [1, 1.05]	2.203 [2.18, 2.24]	1.286 [1.28, 1.29]

503	16	16	16	1.063 [1.05, 1.08]	2.151 [2.14, 2.16]	1.285 [1.28, 1.29]
1009	16	16	16	1.135 [1.12, 1.15]	2.099 [2.09, 2.11]	1.285 [1.28, 1.29]
1511	16	16	16	1.101 [1.08, 1.12]	2.077 [2.07, 2.09]	1.285 [1.28, 1.29]
1823	16	16	16	1.105 [1.08, 1.13]	2.11 [2.1, 2.12]	1.285 [1.28, 1.29]

T5. Exponent of total F_p multiplications in ℓ

quantity	slope [95%]	sizes
sieve F1	−0.152 [−0.208, −0.107]	251–1823
sieve F2	0.042 [−0.020, 0.103]	1009–1823
strong rho, harness	0.511 [0.483, 0.540]	251–1823
strong rho, tuned	0.511 [0.482, 0.540]	251–1823

T6. Decisions (PROTOCOL.md D1–D6)

Withheld: acceptance rules A1-A7 did not all pass. ['A4: rho A/A at p=1009']